

# SANTA CLARA UNIVERSITY

Leavey School of Business

## Natural Language Processing and AI

### ISBA 2411 | Course Assignments

Santa Clara University | Leavey School of Business

### Final Team Project Milestones

All graded coursework outside the midterm is delivered through the Final Team Project, a multi-milestone group project (max 4 students) spanning the full course. Teams build a functional NLP system addressing a real organizational or social problem; each milestone applies that part of the course to your own project, so the work builds cumulatively toward the final system. All milestones are submitted via Camino; the final presentations and demos are delivered live across both Week 10 classes (Aug 18 and Aug 20). Sample projects from previous years are available on Camino.

Topic is open. Any task that fits a baseline-plus-measured-improvement frame is fair game: the capability (classification, retrieval/RAG, summarization, extraction, clustering) is the stable part, while the domain is your team's to choose. The default techniques named in each milestone are starting points, not requirements. Assignments are spaced every other week (Weeks 1, 3, 5, 7, and 9); the final system is due in Week 9 and teams present live across both Week 10 classes (Aug 18 and Aug 20).

#### Assignment 1: Milestone 1 — Business Brief (Week 1)

Frame your proposed final team project as a short business brief: identify a real-world NLP problem, the target user or organization, the value proposition, and an initial hypothesis about which NLP capabilities could address it, together with key risks and governance considerations. 400–600 words. Completed individually and used to form project teams; the problem you define here is the one your team carries through every later milestone.

#### Assignment 2: Milestone 2 — Project Baseline & Representation (Week 3)

Represent your team's own dataset and establish a first quantitative baseline for your problem. Apply static and contextual embeddings to a similarity or clustering task relevant to your project, and evaluate quality using an intrinsic benchmark and a task-specific metric. Where applicable, include a short bias check on at least one representation. Deliverable: notebook + 500-word reflection. Builds toward the final project by giving your team its first measurable footing and a baseline to improve on.

#### Assignment 3: Milestone 3 — Model Adaptation Experiment (Week 5)

Experiment with adapting a model to your team's task and report what you learn. Compare adaptation strategies — zero-shot prompting, few-shot prompting, and LoRA fine-tuning as the default set — on performance, cost-efficiency, and practical tradeoffs. Deliverable: notebook + 500-word analysis of which strategy best fits your project and why. Builds toward the final project by identifying the adaptation approach your system will use, or deliberately rule out.

#### Assignment 4: Milestone 4 — System Prototype (Week 7)

Deliver a working end-to-end prototype of your team's system: a pipeline that takes real input, produces output, and reports a preliminary evaluation against a baseline with at least one metric. Retrieval-augmented generation — a document corpus, embedding and retrieval pipeline, generation layer, grounding/citation mechanism, and a hallucination analysis — is the default path and a natural fit for

Q&A-style projects; a classification, extraction, or summarization pipeline is equally acceptable. Deliverable: GitHub repo link + 2-page technical summary. Builds toward the final project as the first integrated, demonstrable version of your system.

**Assignment 5: Milestone 5 — Final Presentation & Demo (Weeks 9–10)**

Submit your production-ready system — GitHub repo, full documentation, evaluation results (baseline, measured improvement, and a metric), governance/risk appendix, slide deck, and recorded demo video — by the end of Week 9. Each team then presents live in a 10-minute block (about 7–8 minutes presenting plus 2–3 minutes of Q&A); presentations are spread across both Week 10 classes (Aug 18 and Aug 20).